Customer Segmentation Project
K-Means Clustering

Machine Learning Assignment 11

October 7, 2026

1 Introduction

In this assignment, we apply unsupervised machine learning—specifically the K-Means Clustering algorithm—to a customer dataset containing Annual Income and Spending Scores. The goal is to discover hidden patterns in the data and segment customers into distinct groups to drive targeted marketing strategies.

2 Determining the Number of Clusters (Elbow Method)

Before clustering the data, we must determine an appropriate value for k (the number of clusters). We iterate k from 1 to 10, computing the Within-Cluster Sum of Squares (WCSS) for each.

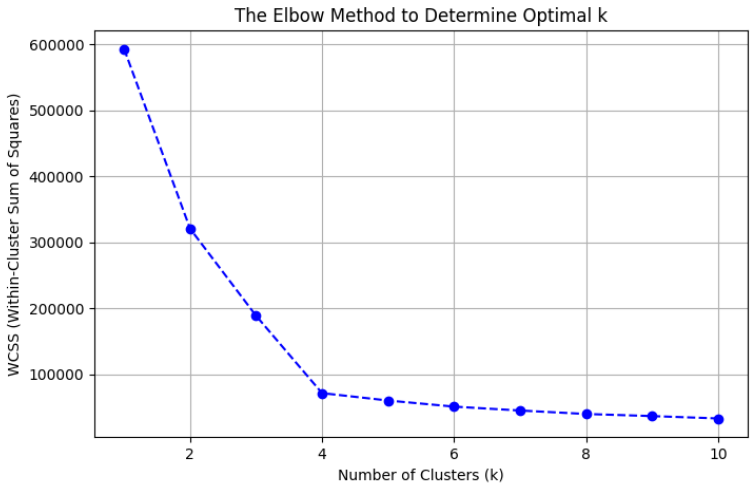


Figure 1: The Elbow Graph showing WCSS for various values of k.

As seen in Figure **??**, the WCSS drops drastically from $k = 1$ to $k = 4$. After $k = 4$, the decrease in WCSS becomes marginal and almost linear. This point of inflection—the "elbow"—indicates that $k = 4$ is the optimal number of clusters for this dataset.

3 Cluster Visualization and Business Interpretation

Applying K-Means with $k = 4$ yields the following segmentation:

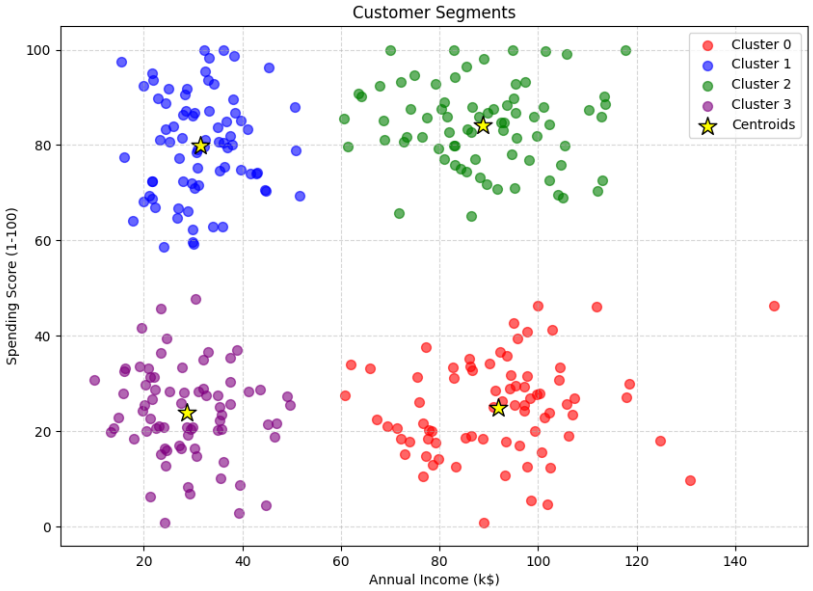


Figure 2: Customer segments mapped by Annual Income and Spending Score.

Business Interpretation of Segments

By analyzing the average income and spending scores of each cluster, a marketing team can derive actionable insights:

- **Cluster 0 (High Income, Low Spending) - "Careful Savers":** These customers have strong financial resources but spend very little. *Marketing Action:* They might be risk-averse or looking for high-value/luxury investments. Targeted campaigns should focus on quality, durability, and premium services rather than discounts.
- **Cluster 1 (Low Income, High Spending) - "Impulsive Buyers":** Despite having lower incomes, these individuals spend heavily. *Marketing Action:* While they are highly responsive to trends, the company must be careful about pushing them into financial distress. Flash sales, discounts, and low-ticket trendy items perform best here.
- **Cluster 2 (High Income, High Spending) - "VIPs / Core Audience":** This is the most lucrative segment. They earn well and spend freely. *Marketing Action:* Enroll them in premium loyalty programs, offer early access to new products, and provide highly personalized concierge-style customer service to maximize retention.
- **Cluster 3 (Low Income, Low Spending) - "Budget Conscious":** These customers have limited disposable income and spend minimally. *Marketing Action:* Do not waste expensive marketing spend on this segment. Broad, low-cost awareness campaigns and essential/value-pack items might attract them on rare occasions.